

**ANALYSING THE IMPACT OF FOREIGN PLAYERS ON TEAM PERFORMANCE
IN THE IPL 2024**

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1.0 INTRODUCTION

The IPL is India's annual men's Twenty20 (T20) cricket league. The BCCI formed it in 2007 and has 10 state or city-based franchise teams (Agur, 2013). The league runs from March to May in the summer and has an exclusive window in the ICC Future visits Programme, reducing international cricket visits during the IPL. This IPL featured prominent international players. International players have a big impact on game results and team performance. This study examines how international players affect IPL 2024 team performance. This analysis examines performance measurements and match results to determine how important international players are to their teams. This study examines how international players affect IPL 2024 team performance. We analyse crucial performance data to determine how much international players help their clubs succeed.

2.0 OBJECTIVES

To evaluate and compare the performance of foreign and local players in IPL 2024 by analysing their performance metrics using probability, descriptive statistics (mean, median, mode), and frequency distribution. The aim is to determine which group performs better overall in the tournament.

3.0 DATA COLLECTION

For this analysis, the following data will be collected:

The data for this report was collected from the official IPL statistics for the top 10 Indian and overseas players of the season. The following statistics were gathered for each player:

- Player Name: The name of the player.
- Team: The team the player represents.
- Matches (MAT): The number of matches played.
- Innings (INNS): The number of innings batted.
- Not Outs (NO): The number of times the player remained not out.
- Runs: Total runs scored by the player.
- Highest Score (HS): The highest score in a single inning.
- Batting Average (AVG): The average number of runs scored per inning.
- Balls Faced (BF): The total number of balls faced.
- Strike Rate (SR): The rate at which runs are scored per 100 balls.
- Hundreds (100): The number of centuries (100+ runs) scored.
- Fifties (50): The number of half-centuries (50+ runs) scored.
- Fours (4s): The total number of boundaries hit.
- Sixes (6s): The total number of sixes hit.

POS	PLAYER	MAT	INNS	NO	RUNS	HS	AVG	BF	SR	100	50	4S	6S
1	 Virat Kohli RCB	15	15	3	741	113*	61.75	479	154.69	1	5	62	38
2	 Ruturaj Gaikwad CSK	14	14	3	583	108*	53.00	413	141.16	1	4	58	18
3	 Riyan Parag RR	16	14	3	573	84*	52.09	384	149.21	0	4	40	33
5	 Sanju Samson RR	16	15	4	531	86	48.27	346	153.46	0	5	48	24
6	 Sai Sudharsan GT	12	12	1	527	103	47.91	373	141.28	1	2	48	16
7	 K L Rahul LSG	14	14	0	520	82	37.14	382	136.12	0	4	45	19
10	 Abhishek Sharma SRH	16	16	1	484	75*	32.27	237	204.21	0	3	36	42
12	 Rishabh Pant DC	13	13	2	446	88*	40.55	287	155.40	0	3	36	25
15	 Yashasvi Jaiswal RR	16	15	1	435	104*	31.07	279	155.91	1	1	54	16
16	 Shubman Gill GT	12	12	1	426	104	38.73	289	147.40	1	2	37	15

Raw data for local players

POS	PLAYER	MAT	INNS	NO	RUNS	HS	AVG	BF	SR	100	50	4S	6S
4	 Travis Head SRH	15	15	1	567	102	40.50	296	191.55	1	4	64	32
8	 Nicholas Pooran LSG	14	14	6	499	75	62.38	280	178.21	0	3	35	36
9	 Sunil Narine KKR	15	14	0	488	109	34.86	270	180.74	1	3	50	33
11	 Heinrich Klaasen SRH	16	15	3	479	80*	39.92	280	171.07	0	4	19	38
13	 Faf Du Plessis RCB	15	15	0	438	64	29.20	271	161.62	0	4	47	21
14	 Phil Salt KKR	12	12	1	435	89*	39.55	239	182.00	0	4	50	24
21	 Marcus Stoinis LSG	14	14	2	388	124*	32.33	263	147.52	1	2	39	16
22	 Tristan Stubbs DC	14	13	6	378	71*	54.00	198	190.90	0	3	24	26
24	 Jos Buttler RR	11	11	2	359	107*	39.89	255	140.78	2	0	36	12
29	 Jake Fraser-McGurk DC	9	9	0	330	84	36.67	141	234.04	0	4	32	28

Raw data for Overseas players

4.0 PRESENTATION OF DATA

4.1 Chapter 1

Question 01: Probability of a player having an average above 40

Answer: Local Players

- Total Indian players: 10
- Players with an average above 40: Virat Kohli (61.75), Ruturaj Gaikwad (53.00), Sanju Samson (48.27), Rishabh Pant (40.55) = 4
- Probability: $4 / 10 = 0.4$ or 40%

Answer: Foreign Players

- Total overseas players: 10
- Players with an average above 40: Travis Head (40.50), Nicholas Pooran (62.38), Heinrich Klaasen (39.92), Phil Salt (39.55), and Jos Buttler (39.89) = 2
- Probability: $2 / 10 = 0.2$ or 20%

Analyse answer - Among the players mentioned, Indian players are twice as likely as foreign players to have a batting average above 40.

Question 02: Probability of a player hitting more than 30 sixes

Answer: Local Players

- Total Indian players: 10

- Players with more than 30 sixes: Virat Kohli (38), Riyan Parag (33), Sanju Samson (24), Abhishek Sharma (42), Rishabh Pant (25) = 3
- Probability: $3 / 10 = 0.3$ or 30%

Answer: Foreign Players

- Total overseas players: 10
- Players with more than 30 sixes: Travis Head (32), Nicholas Pooran (36), Sunil Narine (33), Heinrich Klaasen (38), Jake Fraser-McGurk (28) = 4
- Probability: $4 / 10 = 0.4$ or 40%

Analyse answer - Among the players mentioned, foreign players are 10% more likely than local players to hit more than 30 sixes.

Question 03: Probability of a player having a strike rate above 150

Answer: Local Players

- Total Indian players: 10
- Players with SR > 150: Virat Kohli (154.69), Sanju Samson (153.46), Abhishek Sharma (204.21), Rishabh Pant (155.40), Yashasvi Jaiswal (155.91), Shubman Gill (147.40) = 5
- Probability: $5 / 10 = 0.5$ or 50%

Answer: Foreign Players

- Total overseas players: 10

- Players with $SR > 150$: Travis Head (191.55), Nicholas Pooran (178.21), Sunil Narine (180.74), Heinrich Klaasen (171.07), Phil Salt (182.00), Tristan Stubbs (190.90), Jake Fraser-McGurk (234.04) = 7
- Probability: $7 / 10 = 0.7$ or 70%

Analyse answer - Among the players mentioned, foreign players are 20% more likely than local players to have a strike rate above 150.

4.2 Chapter 6

For Indian (Local Players) players:

First, sort the average if the each Indian players in ascending order:

31.07, 32.27, 37.14, 38.73, 40.55, 47.91, 48.27, 52.09, 53.00, 61.75

Class Interval	Frequency	Cumulative Frequency	Lower Boundary	Higher Boundary
30.0 - 36.6	2	2	29.5	36.65
36.7 - 43.3	3	5	36.65	43.35
43.4 - 50.0	2	7	43.35	50.05
50.1 - 56.7	2	9	50.05	56.75
56.8 - 63.0	1	10	56.75	63.05

Values for the Median Class:

$L = 36.65$ (lower boundary of the median class)

$N = 10$ (total number of frequencies)

$F = 2$ (cumulative frequency of the class before the median class)

$f = 3$ (frequency of the median class)

$C = 43.35 - 36.65 = 6.7$ (class width)

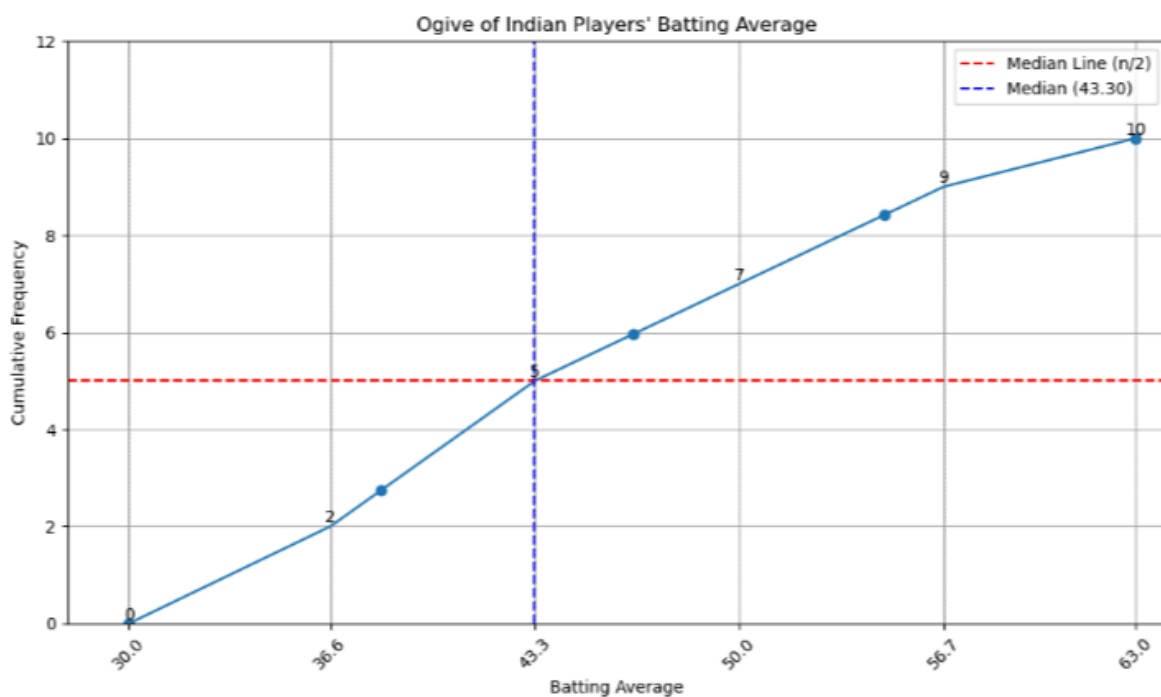
Substituting the values into the formula:

Median= $36.65 + ((10/2)-2 / 3) \times 6.7$

Median = $36.65 + (3 / 3) \times 6.7$

Median = $36.65 + 6.7 = 43$

So, the **Median** batting average is **43.35**.



For Foreign Players :

First, sort the average if the each Overseas players in ascending order:

29.20, 32.33, 34.86, 36.67, 39.55, 39.89, 39.92, 40.50, 54.00, 62.38

Class Interval	Frequency	Cumulative Frequency	Lower Boundary	Upper Boundary
29.1 - 35.8	3	3	29.05	35.85
35.9 - 42.6	5	8	35.85	42.65
42.7 - 49.4	0	8	42.65	49.45
49.5 - 56.2	1	9	49.45	56.25
56.3 - 63.0	1	10	56.25	63.05

$L = 35.85$ (lower boundary of the median class)

$N = 10$ (total number of frequencies)

$F = 3$ (cumulative frequency of the class before the median class)

$f = 5$ (frequency of the median class)

$W = 35.85 - 29.05 = 6.8$ (class width)

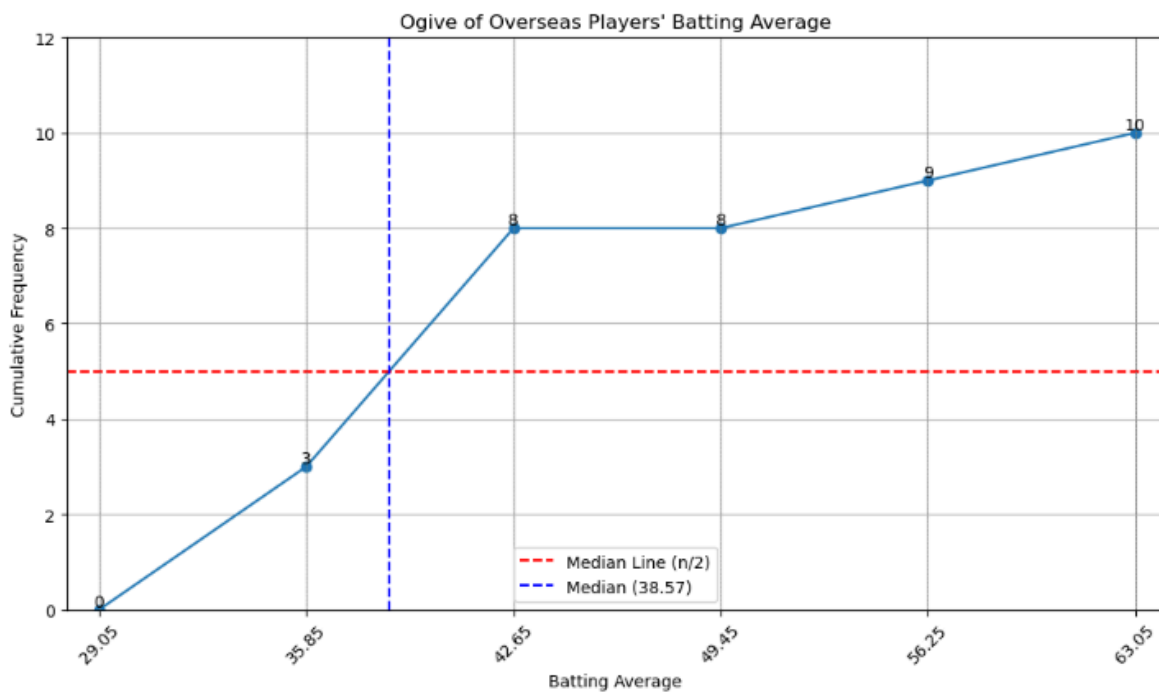
Substituting the values into the formula:

$$\text{Median} = 35.85 + \left(\left(\frac{10}{2} \right) - \frac{2}{5} \right) \times 6.8$$

$$\text{Median} = 35.85 + \left(\frac{2}{5} \right) \times 6.8$$

$$\text{Median} = 35.85 + 2.72 = 38.57$$

So, the **Median** batting average is **38.57**



Analyse answer -

For local players, the median batting average is 43.35, while for foreign players, it's 38.57.

This indicates that, in the given data, local players have a higher median batting average compared to foreign players.

4.3 Chapter 7

4.3.1 MEAN :

For getting the mean, we have take the data of the highest run from each local and foreign player:

To get the mean (average) of the highest runs scored by each local player in the IPL match, we need to sum all the scores and then divide by the number of players.

1. Sum of the highest runs: $113+108+84+86+103+82+75+88+104+104=947$
2. Number of players: 10

Mean = Sum of the highest run / number of player

$$= 947/10$$

$$= 94.7$$

So, the **mean** of the highest runs scored by each local player is **94.7**.

MEAN Of Foreign Players (using highest runs)

1. Sum the highest runs: $102 + 75+109+80+64+89+124+71+107+84= 905$
2. the number of players: 10

Mean = Sum of the highest number / number of player

$$= 905/10$$

$$= 90.5$$

So, the **mean** of the highest runs scored by each foreign local player is **90.5**.

Analysis According to MEAN :

1. Comparison of Means:

- The mean highest runs for local players is 94.7.
- The mean highest runs for foreign players is 90.5.

2. Performance Insights:

- The local players have a higher mean score (94.7) compared to the foreign players (90.5), indicating that on average, local players have performed slightly better in terms of their highest individual scores.
- The difference between the means is $94.7 - 90.5 = 4.2$ which means that the local player's performance is slightly better than the foreign player.

4.3.2 MODE :

MODE of Local Players by using strike rate :

We use the strike rate value of each local player in IPL 2024 as class interval and frequency :

Class interval	frequency	Class boundary
136.12 - 149.74	5	136.115 - 149.745
149.74 - 163.36	4	149.735 - 163.365
163.36 - 176.98	0	163.355 - 176.985
176.98 - 190.60	0	176.975 - 190.605
190.60 - 204.21	1	190.595 - 204.215

The data in the table is grouped data. To get the mode for the grouped data, we need to use the formula for the mode in a frequency distribution. The mode for grouped data is calculated using the following formula:

$$\text{Mode} : L + \left(\frac{f_m - f_{m-1}}{(f_m - f_{m-1}) + (f_m - f_{m+1})} \right) * h$$

Modal class: 136.12 - 149.74

$L = 136.12$ (lower boundary of the modal class.)

$f_m = 5$ (frequency of the modal class)

$f_{m-1} = 0$ (frequency of the class before the modal class)

$f_{m+1} = 4$ (frequency of the class after the modal class)

$h = 13.62$ (class width)

$$\text{Mode} = 136.12 + \left(\frac{(5-0)}{(5-0)+(5-4)} \right) \times 13.62$$

$$= 136.12 + 0.8333 \times 13.62$$

$$\approx 147.47$$

The **mode** for the local player is **147.47**

Mode of Foreign Player by using strike rate :

Class Interval	frequency	Class boundary
147.52 - 164.82	147.515 - 164.825	1
164.82 - 182.12	164.815 - 182.125	2
182.12 - 199.42	182.115 - 199.425	3
199.42 - 216.72	199.415 - 216.725	2
216.72 - 234.04	216.715 - 234.045	1

Here

$L=182.12-199.42$ (lower boundary of the modal class)

$f_m=3$ (frequency of the modal class)

$f_{m-1}=2$ (frequency of the class before the modal class)

$f_{m+1}=2$ (frequency of the class after the modal class)

$h=17.30$ (class width)

Mode :

$$\begin{aligned} &= 182.12 + ((3-2)/(3-2)+(3-2)) \times 17.30 \\ &= 182.12 + 8.65 \\ &= 190.77 \end{aligned}$$

Performance Analysis Based on MODE :

The mode value for local players is 147.47, whereas for foreign players, it is significantly higher at 190.77. foreign players frequently achieve their highest strike rates around 190.77. This much higher mode value indicates that foreign players are capable of delivering exceptionally high-performance innings in IPL 2024. Therefore, based on the mode analysis, foreign players are better performers in the context of achieving high strike rates in IPL matches.

4.3.3 MEDIAN :

Median for local player (using highest runs):

Class interval	Class Boundary	Midpoint	Frequency	Cumulative Frequency
75 - 82	74.5 - 82.5	78.5	2	2
83 - 90	82.5 - 90.5	86.5	3	5
91 - 98	90.5 - 98.5	94.5	0	5
99 - 106	98.5 - 106.5	102.5	3	8
107 - 114	106.5 - 114.5	110.5	2	10

To get the median value for local player,

$$\text{Median} = L + (N / 2 - CF / f) \times c$$

Where:

$N = 10$ (total number of observations)

So the position of the median class is : $N/2 = 10/2 = 5$ th class that is (83-90)

$L = 82.5$ (lower boundary of the median class)

$CF = 2$ (cumulative frequency of the class preceding the median class)

$f = 3$ (frequency of the median class)

$c = 7$ (class size)

$$\text{Median} = 82.5 + (5 - 2/3) \times 7$$

$$= 82.5 + 1 \times 7$$

$$= 89.5$$

So the **median** value for the Indian local player is **89.5**,

Median value of the foreign player (using highest runs):

Class Interval	Class Boundary	Midpoint	Frequency	Cumulative Frequency
64 - 75	112 - 124	69.5	3	3
76 - 87	75.5 - 87.5	81.5	2	5
88 - 99	87.5 - 99.5	93.5	1	6
100 - 111	99.5 - 111.5	105.5	3	9
112 - 123	111.5 - 123.5	117.5	1	10

To get the median value for foreign players,

$$\text{Median} = L + (N / 2 - CF / f) \times c$$

Where:

$N = 10$ (total number of observation)

So the position of the median class is : $N/2 = 10/2 = 5$ th class that is (76-87)

$L = 75.5$ (Lower boundary)

$CF = 3$ (Cumulative frequency of the class preceding the median class)

$f = 2$ (Frequency of the median class)

$c = 11$ (Class size)

$$\begin{aligned}\text{Median} &= L + (N / 2 - CF / f) \times c \\ &= 75.5 + (5 - 3 / 2) \times 11 \\ &= 75.5 + 1 \times 11 \\ &= 86.5\end{aligned}$$

So the **Median** for Foreign players in the IPL is **86.5**

Analysis based on the median of both local and foreign players in the IPL :

- The median run score for the local player (89.5) is higher than that for the foreign player (86.5).
- The median represents the middle value in a data set when it is ordered from least to greatest. A higher median indicates that the central tendency of the local player's performance is better than that of the foreign players

5.0 ROLES AND RESPONSIBILITIES OF PROJECT MEMBERS

- MD Parvej Ahmed Rafi
 - Role: Data Collection and Data Presentation
 - Responsibilities:
 - Apply key statistical concepts such as Data Description (Mean, Mode , and median) to analyse the IPL player data.
- Mubarak Mohamed Hussein Liban
 - Role: Report Writing (Introduction, Objectives, and Methodology)
 - Responsibilities:
 - Describe the methodology used for data collection and analysis, ensuring it is detailed and logical.
- Muhammad Amirul Mustaqin Bin Abdul Mutalib
 - Role: Data Collection, Report Making, and Overall Checking
 - Responsibilities:
 - Contribute to the writing of various sections of the report, ensuring clarity and coherence.
- MD Shakib Mostafa
 - Role: Data Collection, Report Making, and Calculation Chapter
 - Responsibilities:
 - Participate in data collection efforts, focusing on obtaining accurate and relevant data.

- Mohemedh Sharaf Sahir
 - Role: Data Collection, Report Making, and Calculation
 - Responsibilities:
 - Apply key statistical concepts such as Probability and Frequency Distribution to analyse the IPL player data
 - Create graphs and visual representations of the data to support the analysis.

6.0 CONCLUSION

In analysing IPL 2024 performance statistics, we observed notable trends. Foreign players are 20% more likely to have a strike rate above 150, and their average strike rate peaks at 190.77, indicating exceptional high-performance potential. Conversely, local players exhibited a higher median batting average (43.35 vs. 38.57), mean score (94.7 vs. 90.5), and median run score (89.5 vs. 86.5), suggesting superior consistency and overall performance. Despite foreign players' explosive potential, local players showed stronger central tendencies and slightly better average performance, making them more reliable contributors to team success. Overcoming data collection and mathematical modelling challenges, we recommend team owners prioritise high-quality foreign players and retain top performers to enhance team competitiveness and success.

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8.0 Turnitin Report

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